

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-9577
Project Name	Import Data using Transform Maps (Spreadsheet)SWTID-2026-9577
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Project: Import Data using Transform Maps (Spreadsheet) - TeamOverviewWorkspaceKanban

Flow: (1) Employee data is prepared in an Excel file. (2) The file is attached to a Data Source and loaded. (3) Rows are stored in the Import Set staging table. (4) A Transform Map maps source fields to target fields. (5) Records are created in the target table and verified using Import Set Rows and Transform History. (6) The data is shown in the TeamOverview Workspace Kanban view.

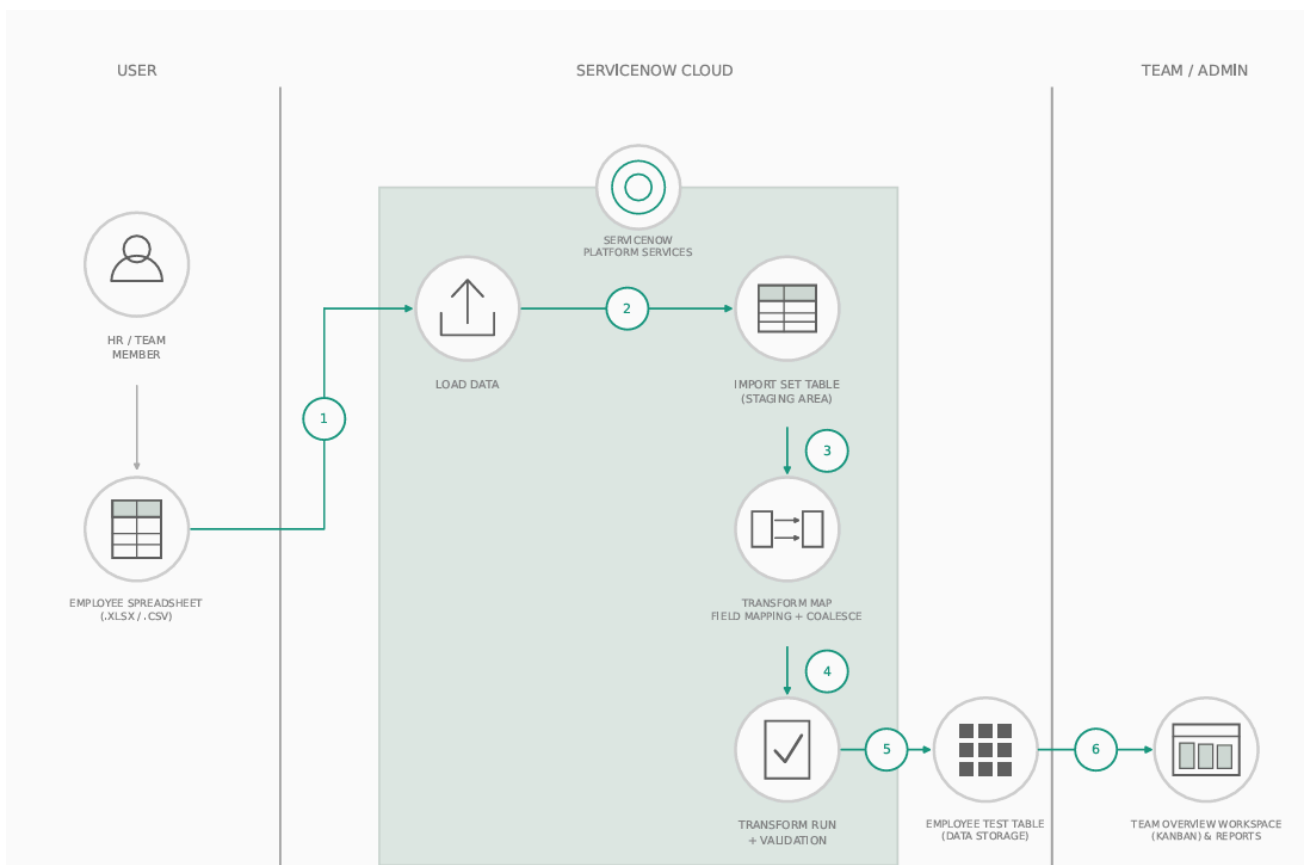


Figure 1: Architecture - importing Excel data into ServiceNow using Import Set and Transform Map

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application: ServiceNow platform UI for loading data and a Kanban-style team overview in a workspace.	ServiceNow Next Experience UI, Workspace (UI Builder), HTML, CSS, JavaScript
2	Application Logic-1	Import Set and Transform Map: loads spreadsheet rows into the staging table and maps source fields to target table fields.	ServiceNow Import Sets, Transform Maps, Field Maps
3	Application Logic-2	Data cleansing and validation during transformation (coalesce on a unique field, field transformations, onBefore / onAfter scripts).	Server-side JavaScript (Transform Scripts, Glide API)
4	Application Logic-3	Kanban view of the imported team data, grouped by a status / team field for a team overview.	Visual Task Board / Workspace Kanban, UI Builder
5	Database	Import Set (staging) table holds raw spreadsheet data; the target table holds the final employee records.	ServiceNow relational tables (u_import set table, sys_user / custom table)
6	Cloud Database	Database service hosted on the cloud, managed by the ServiceNow platform.	ServiceNow Cloud Instance (Personal Developer Instance)
7	File Storage	The Excel file is stored as an attachment on the Data Source record.	ServiceNow Attachments (sys_attachment)
8	External API-1	Not used. The data is loaded manually from a spreadsheet file.	N/A (optionally REST Table API)
9	External API-2	Not used.	N/A
10	Machine Learning Model	Not used in this project.	N/A
11	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: not applicable. Cloud Server Configuration: ServiceNow PDI (developer.servicenow.com), browser access.	ServiceNow Cloud (PDI)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks used. The project mainly uses the ServiceNow low-code platform; standard web technologies are used for the UI.	JavaScript (Rhino engine in ServiceNow), HTML, CSS
2	Security Implementations	Access to import tables, transform maps and target data is restricted by roles; data is transferred over encrypted connections.	Role-based ACLs, IAM / user roles, HTTPS (TLS), Data policies

3	Scalable Architecture	Multi-tier architecture: UI tier, application tier (Import Set / Transform Map) and database tier. Staging tables allow large bulk imports to be processed in batches.	ServiceNow multi-tier cloud platform
4	Availability	Hosted on ServiceNow cloud infrastructure, which provides managed hosting, backups and failover.	ServiceNow Cloud (managed instance)

S.No	Characteristics	Description	Technology
5	Performance	Bulk data is loaded into a staging table first, then transformed in one run; coalesce prevents duplicate records and keeps imports efficient.	Import Set staging table, Transform Map coalesce, indexed fields

References:

<https://docs.servicenow.com/>

<https://developer.servicenow.com/>

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>